# MedLens

## AI Medication Plain-Language Translator

### Product Requirements Document

Version 1.0 | April 2026 | Team 1

Platform: Mobile App (iOS & Android) | Industry: Pharma Tech / Health

DeepSeek API • OpenFDA API • Supabase • React Native

*CONFIDENTIAL — CAPSTONE PROJECT DOCUMENT*

# Executive Summary

MedLens is a mobile application — available on iOS and Android — that translates complex medication information into clear, plain everyday language that anyone can understand. The app is built for patients, caregivers, and everyday people who are handed a prescription or medication leaflet and struggle to make sense of what it says.

Medication leaflets and prescription documents are written in dense medical jargon designed for healthcare professionals, not for the patients who actually take the drugs. A patient who picks up a prescription and cannot understand the dosage instructions, warnings, or side effects is at genuine risk of taking their medication incorrectly. This is the daily experience of the majority of patients worldwide, and it is even more acute in regions where English is a second language.

MedLens addresses this directly. A user opens the app, types the name of a medication, and the app retrieves official drug data from the OpenFDA API. That data is passed to DeepSeek AI, which rewrites the medical information in simple, everyday language — what the drug is for, how to take it, what warnings to be aware of, and what side effects are possible. Beyond the core explanation, users can save medications to a personal medicine cabinet, check for risky drug combinations in plain language, export a summary card to share with a caregiver, and activate an 'Explain Like I'm 12' toggle for even simpler language.

The app runs on a fully free technology stack — OpenFDA for drug data, DeepSeek API for AI-powered rewriting, and Supabase for user accounts and saved cabinets — built with React Native, the same framework that powers the ChatGPT mobile app. MedLens is not a diagnostic tool. It is a health literacy tool that helps people understand the medications they have already been prescribed, so they can take them correctly, safely, and with confidence.

# Problem Statement

Medication leaflets and prescription information are written in dense medical jargon. Patients routinely misunderstand dosage instructions, side effects, and drug interactions — especially in regions where English is a second language. Key problems include:

- **Low Health Literacy:** Many users cannot correctly interpret dosage instructions, warnings, or side effects from standard prescription labels or medication leaflets.
- **Medication Misuse:** Misunderstanding instructions leads to incorrect usage, missed doses, or dangerous over-use — all of which carry serious health consequences.
- **Language Barriers:** English-only medical information limits accessibility in multilingual regions, particularly across West Africa where English is a second language for the majority of the population.
- **Limited Accessibility:** Users with lower literacy levels or visual impairments face additional barriers in reading and understanding written medication instructions.

As a result, patients routinely make unsafe or uninformed decisions about their medications — not because they are careless, but because the information they are given was never written for them.

# Product Vision and Goals

## Vision

To become a trusted mobile platform that helps people understand their medications clearly, safely, and confidently — in language that works for them.

## Core Goals

- **Clarity:** Translate medical jargon into simple, human-friendly language anyone can understand.
- **Safety:** Prevent harmful misinterpretation of medication instructions by providing accurate, plain-language explanations grounded in verified drug data.
- **Accessibility:** Support voice playback and simplified language modes so no user is left behind, regardless of literacy level or language background.
- **Trust:** Provide transparent, responsibly simplified information — always clearly sourced from official drug data, never invented by AI.
- **Empowerment:** Enable users to make informed decisions about their medications and have better conversations with their healthcare providers.

# Target Users and Use Cases

## Primary Target Users

- Adults managing their own prescriptions
- Caregivers assisting elderly parents, children, or family members with medications
- Individuals with moderate literacy challenges
- Non-native English speakers in multilingual regions

## Key Use Cases

### Use Case 1 — Understanding a Prescription

A user receives a new prescription and cannot understand the label. They open MedLens, type the medication name, and receive a plain-language explanation of what the drug is for, how to take it, and what to watch for.

### Use Case 2 — Medication Management

A user saves multiple medications to their personal medicine cabinet. Each time they need a reminder of how to take a specific drug, they open the cabinet and review the saved plain-language summary without re-searching.

### Use Case 3 — Basic Interaction Awareness

A user is prescribed a new medication and wants to know if it is safe to take alongside their existing drugs. They select both medications in the interaction checker and receive a plain-language flag if a known risky combination is detected.

### Use Case 4 — Caregiver Support

A caregiver managing medications for an elderly parent generates an exportable summary card for one of the parent's prescriptions and shares it via WhatsApp with another family member helping with care.

### Use Case 5 — Language Accessibility

A user who finds the standard plain-language summary still too complex activates the 'Explain Like I'm 12' toggle and receives an even simpler version of the same information — shorter sentences, no technical vocabulary, and relatable everyday language.

# Key Features (MVP)

The Minimum Viable Product is defined by five core features, taken directly from the project brief. These are the committed features for the first public release of the MedLens mobile app.

## 1. Drug Search with Autocomplete Suggestions

Users can search for any medication by typing its name into the search bar. The app displays autocomplete suggestions as the user types, pulling from the OpenFDA Drug API in real time. The search supports both brand names and generic names, and handles common spelling variations. Once a medication is selected, the app retrieves the official drug label data and passes it to the DeepSeek AI layer, which rewrites the information in plain, everyday language and displays it as a structured summary card.

## 2. "Explain Like I'm 12" Toggle

Every drug summary screen includes an 'Explain Like I'm 12' toggle. When activated, DeepSeek re-processes the current drug's information using an even simpler prompt — shorter sentences, no technical vocabulary, and everyday analogies where helpful. The toggle is clearly labelled so users understand it provides an additional simplification layer, not a different or reduced version of the medical facts.

This feature is designed for users who find the standard plain-language summary still too complex, including elderly users, low-literacy users, and users reading in their second language.

### 3. Personal Medicine Cabinet

Users can save any medication they have searched to their personal medicine cabinet — a persistent list of their medications stored in their Supabase account. The medicine cabinet is accessible from the home screen and allows users to view the full plain-language summary of any saved medication at any time without re-searching.

The cabinet stores the drug name and its simplified explanation. It does not track dosage schedules or make dosage recommendations in MVP.

### 4. Interaction Checker

Users can select multiple medications from their medicine cabinet and run an interaction check. The system displays interaction warnings only when reliable data is available from OpenFDA. If data is unavailable or inconclusive, the app clearly communicates this:

"We cannot confirm interactions between these medications. Please consult a healthcare professional."

Interaction results are displayed in plain language — not clinical severity labels. The checker does not guess or infer. It only surfaces interactions supported by verified data.

### 5. Exportable Summary Card

After viewing any drug summary, users can generate an exportable summary card. The card includes the drug name, what it does, how to take it, and key warnings — formatted cleanly for sharing. Users can share the card with a caregiver or family member directly from the app using the native iOS and Android share sheet (WhatsApp, email, or any installed messaging app).

## Future Features and Roadmap

As MedLens grows beyond MVP, the following capabilities are planned:

- **Medication Label Scanning (OCR):** A future release will introduce the ability to snap a photo of a medication label or leaflet. The app will extract the drug name from the image and run the same plain-language explanation flow — removing the need to type the medication name manually.
- **Expanded Language Support:** Additional global languages will be added in post-MVP releases, building on the English support available at launch.
- **Advanced Interaction Checker:** Integration with additional verified pharmaceutical datasets to increase the coverage and reliability of interaction results.
- **Medication Reminders:** A scheduling feature that allows users to set daily reminders for each medication in their cabinet.
- **AI Conversational Assistant:** A follow-up Q&A feature where users can ask questions about a specific drug after viewing its summary — powered by DeepSeek with drug-specific context.
- **Nigerian Drug Coverage:** Integration with EMDEX API or DrugList.ng to cover NAFDAC-registered Nigerian medications not found in the OpenFDA database, using RxNorm as a drug name normalisation layer across both sources.

# Free vs. Paid Feature Tiers

MedLens will adopt a freemium model. The core plain-language translation feature is always free. Premium tiers unlock additional functionality for power users.

| Feature | Free | Pro — $5/mo |
|---|---|---|
| Drug search & plain-language summary | ☑ Unlimited | ☑ Unlimited |
| 'Explain Like I'm 12' toggle | ☑ Included | ☑ Included |
| Medicine cabinet (saved medications) | Up to 5 drugs | Unlimited |
| Interaction checker | 2 drugs | Up to 10 drugs |
| Exportable summary card | ☑ Included | ☑ Included |
| Medication reminders (V2) | ✕ | ☑ Included |
| Support | Self-serve | Email priority |

# Detailed User Flows

## Flow 1 — Onboarding & Authentication

1. User downloads and opens MedLens for the first time. Splash screen displays briefly before the onboarding screens load.
2. Onboarding screens walk the user through what MedLens does — three simple slides: search a medication, get a plain-language explanation, save to your cabinet.
3. After the final onboarding slide, a brief loading spinner plays and the user is taken directly to the home screen. No sign-up decision is required upfront.
4. User is already in guest mode and can immediately start searching for a medication and viewing plain-language explanations.
5. When the user attempts to save a medication to their cabinet, they are prompted to create a free account or log in. This is the only moment sign-up is introduced.
6. Returning users who have already signed up bypass onboarding and go directly to the log-in screen. On successful login, they land on their home screen with their cabinet restored.

## Flow 2 — Medication Explanation

1. User opens MedLens. Home screen displays a search bar and their medicine cabinet.
2. User types a medication name. Autocomplete suggestions appear from OpenFDA as they type.
3. User selects the medication from suggestions.
4. App retrieves the official FDA label data from OpenFDA API.
5. Backend passes the label data to DeepSeek API with a plain-language prompt.
6. AI returns a structured plain-language summary: what it is, how to take it, warnings, side effects.
7. User reads the summary. They can toggle 'Explain Like I'm 12' for a further simplified version.
8. User taps 'Save to Cabinet' to add the medication to their personal medicine cabinet.

## Flow 3 — Medicine Cabinet & Interaction Check

1. User opens their medicine cabinet from the home screen.
2. Cabinet displays all saved medications with drug name and one-line summary.
3. User taps 'Check Interactions.' They select two or more medications from the cabinet.
4. App queries OpenFDA for known interaction data between the selected drugs.
5. Results display as a plain-language flag: safe, use caution, or consult a professional.

## Flow 4 — Export & Share Summary Card

1. User views a drug summary screen.
2. User taps 'Export Summary.' App generates a formatted summary card.
3. Card includes: drug name, what it does, how to take it, key warnings.
4. User taps 'Share.' Native iOS/Android share sheet opens.
5. User selects WhatsApp, email, or any installed app to send the card.

# Technical Requirements

## Architecture Overview

MedLens is a mobile application built with React Native — the same framework used to build the ChatGPT mobile app. React Native produces a fully native iOS and Android application from a single codebase, delivering the smooth screen transitions, native keyboard behaviour, and fluid scroll physics that define a professional mobile app experience. The app is distributed via the Apple App Store and Google Play Store.

A branded splash screen — displaying the MedLens logo on a teal background — appears on every app launch using the @notifee/react-native and react-native-splash-screen libraries, configured natively for both platforms.

The architecture separates concerns cleanly across four layers: the React Native mobile frontend, a lightweight Node.js/Express backend API, external data and AI services, and the Supabase database.

## Tech Stack

| Layer | Technology | Role |
|---|---|---|
| **Mobile Frontend** | React Native (Expo) | Builds the native iOS and Android app from a single codebase. Delivers ChatGPT-like smooth navigation — no page reloads, instant screen transitions, native feel throughout. |
| **Splash Screen** | react-native-splash-screen | Branded splash screen shown on every app launch — MedLens logo on teal background — before the React Native UI loads. |
| **Backend** | Node.js + Express | Lightweight API server. Orchestrates calls to OpenFDA and DeepSeek, handles Supabase auth, and returns structured responses to the mobile app. |
| **AI Engine** | DeepSeek API (free tier) | Rewrites FDA label data into plain-language summaries. Handles both the standard explanation and the 'Explain Like I'm 12' mode via separate prompt configurations. |
| **Drug Data** | OpenFDA Drug API | Free, no API key required for basic use. Provides official FDA drug label data — |

| | | |
|---|---|---|
| | | indications, dosage, warnings, side effects, and adverse events — for all FDA-approved medications. |
| **Database & Auth** | Supabase (free tier) | PostgreSQL database for user accounts and saved medicine cabinets. Supabase Auth handles sign-up, login, and Google OAuth. Free tier supports up to 50,000 monthly active users. |
| **Export / Share** | react-native-view-shot + Share API | Generates the exportable summary card as an image or PDF. Triggers the native iOS/Android share sheet for sending via WhatsApp, email, or any installed app. |

## API Integrations

| Endpoint | Method | Description |
|---|---|---|
| api.fda.gov/drug/label.json | GET | Retrieve official drug label data by brand or generic name. Returns indications, dosage, warnings, adverse reactions. |
| api.fda.gov/drug/event.json | GET | Retrieve adverse event data used to support the interaction checker. |
| DeepSeek /chat/completions | POST | Send structured FDA label content. Receive plain-language summary JSON for standard mode and ELI12 mode. |
| Supabase /auth/v1/signup | POST | Create a new user account via Supabase Auth. |
| Supabase /rest/v1/cabinet | GET / POST / DELETE | Read and write the authenticated user's saved medicine cabinet. |

## Security and Compliance

- **Data in Transit:** All API communication encrypted over HTTPS/TLS. Supabase enforces SSL on all database connections.
- **Authentication:** Supabase Auth manages JWT-based sessions. Supports email/password and Google OAuth sign-in.
- **Data Minimization:** MedLens stores only the drug names users explicitly save. No sensitive medical records are stored beyond user inputs. No data is sold or shared with third parties.
- **AI Guardrails:** DeepSeek is instructed via system prompt to only rewrite provided FDA label data — never to generate new medical instructions, guess missing information, or recommend treatments. If reliable data is unavailable, the app displays: 'We could not find reliable information for this medication. Please consult a healthcare professional.'

- **App Store Compliance:** Both Apple App Store and Google Play Store submissions include a Privacy Policy, Terms of Service, and health content disclaimer. MedLens is categorised as a Health & Fitness utility — not a regulated medical device — at MVP stage.
- **Disclaimer:** Displayed prominently on every summary screen: 'MedLens simplifies medical information for understanding. It does not replace professional medical advice.'

# Metrics of Success

## Product Metrics

- **Number of Active Users (DAU/MAU):** Daily and monthly active users — the primary indicator of adoption and sustained engagement.
- **Number of Searches per User:** Average drug searches per active user per session — measures genuine utility.
- **Feature Usage:** Usage rates for ELI12 toggle, voice playback, cabinet saves, interaction checker, and export card — identifies which features deliver the most value.
- **Onboarding Completion Rate:** Percentage of sign-ups that complete account setup and run their first drug search.

## Trust Metrics

- **Explanation Helpfulness Rating:** In-app 'Was this helpful? Yes / No' prompt shown after each summary. Target: above 80% positive.
- **Successful Explanation Rate:** Percentage of searches that return a full plain-language summary vs. a 'not found' result.
- **Frequency of 'Data Not Found' Cases:** Tracks coverage gaps in the OpenFDA database — informs the prioritisation of additional data sources.

## Accessibility Metrics

- **ELI12 Toggle Usage Rate:** How often users activate the simplified mode — signals when standard plain language is still too complex.
- **Text-to-Speech Usage Rate:** Percentage of users who activate voice playback — indicates the size of the low-literacy and visually impaired user segment.
- **App Store Rating:** Target 4.5 / 5 stars or above on both the Apple App Store and Google Play Store within 3 months of launch.

# Risks and Limitations

- **AI Simplification Nuance:** AI may occasionally miss nuance or simplify a warning in a way that reduces its urgency. Mitigated by restricting DeepSeek to only rewrite provided data and by displaying the FDA source alongside every summary.
- **OpenFDA Data Coverage:** OpenFDA covers FDA-approved medications and does not include non-US drug brands. Mitigated in MVP by clearly communicating this limitation; broader global drug coverage is on the roadmap.
- **User Over-Reliance:** Some users may treat MedLens summaries as a substitute for professional medical advice. Mitigated by prominent disclaimers on every screen and consistent 'consult your healthcare provider' messaging at key moments.
- **Interaction Checker Coverage:** The interaction checker only surfaces interactions present in OpenFDA data. If data is unavailable, the app communicates this clearly rather than guessing.

### MVP Limitations

- English only at launch. Additional global languages are planned for post-MVP releases.
- Medicine cabinet stores drug name and simplified explanation only. Dosage tracking and scheduling are not included in MVP.
- Interaction checker limited to data available in OpenFDA adverse event records. It is not a comprehensive clinical pharmacology tool.

## Competitive Landscape and Analysis

MedLens operates at the intersection of consumer health literacy and AI-powered medical communication. The competitors below are evaluated against MedLens's core offering: a free, direct-to-patient mobile app that translates medication information into plain language using official FDA data.

### Key Competitors

#### Vital.io

An AI-powered tool that translates clinical notes and lab results into plain language. Hospital-facing only — not available directly to patients. Does not focus on medication labels specifically.

#### Patiently AI

A UK-based tool that translates medical letters and clinical documents into everyday language. Focused on documents and letters, not medication name search. UK/NHS-centric — not FDA-powered.

#### drugs.com / WebMD

The incumbent consumer drug information platforms. Comprehensive drug databases but present information in clinical format, are ad-heavy, and have no AI plain-language simplification layer. Their side-effect pages are a primary driver of patient health anxiety online.

### GoodRx

A widely used drug pricing and discount platform with basic drug information pages. Focused on cost savings, not health literacy. No AI simplification layer.

## Competitive Matrix

| Feature | MedLens | Vital.io | Patiently AI | drugs.com / WebMD | GoodRx |
|---|---|---|---|---|---|
| Plain-language simplification | ✓ | ✓ | ✓ | ✕ | ✕ |
| Medication-specific focus | ✓ | Partial | ✕ | ✓ | Partial |
| Free & consumer-direct | ✓ | ✕ | Partial | ✓ | ✓ |
| Official FDA data source | ✓ | ✕ | ✕ | Partial | ✕ |
| AI-generated summaries | ✓ | ✓ | ✓ | ✕ | ✕ |
| Mobile app (iOS & Android) | ✓ | ✕ | ✕ | Partial | ✓ |
| Ad-free experience | ✓ | ✓ | ✓ | ✕ | Partial |
| Drug interaction checker | ✓ | ✕ | ✕ | ✓ | ✕ |
| Exportable summary card | ✓ | ✕ | ✕ | ✕ | ✕ |

MedLens is the only free, consumer-facing, FDA-powered, AI plain-language medication translator built as a native mobile app. Every competitor either requires institutional access, focuses on a different input type, presents clinical data without simplification, or is not available as a direct-to-patient mobile experience.